Pablo Martín Palomino

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Summary

- Computer Science and Mathematics graduate (double degree, University of Granada), specialized in Artificial Intelligence and now in the MSc in Data Science and Computer Engineering. I build complete, tested software backed by a solid mathematical and algorithmic foundation. I have hands-on experience building and fine-tuning neural networks, and interests across AI applications, software development and game development. Seeking a software engineering or AI/ML internship or junior role.

Education

- 2021–2026 **Double Bachelor's Degree in Computer Science and Mathematics**, University of Granada
  Specialization in Artificial Intelligence
  Thesis: *From the Theory of f-Divergences to Practice: Implementation and Evaluation of GANs*
  Implemented and compared 7 GAN variants (from vanilla GAN to WGAN-GP and a CycleGAN) in PyTorch on 5 datasets, evaluated with FID and IS over up to 10 runs.
- 2026–present **MSc in Data Science and Computer Engineering**, University of Granada
- Google Cloud Data Analytics Certificate, Google Cloud (2025)

Projects

- 2026 **Voice banking assistant (agentic LLM).** Built for the Unicaja & UGR Responsible AI Challenge as main contributor in a team of 3. Fully local on Qwen3-8B (Ollama): tool-calling agent, read-only text-to-SQL with self-correction, and month-end spend forecasting. FastAPI + WebSocket backend with atomic money transfers and 33 deterministic tests. 96.8% tool choice and 90.5% answer accuracy on a 44-question eval harness.
- **sam2-lite: efficient video segmentation.** Made SAM 2.1 video tracking 2.1× faster on CPU (J&F 84.4 vs. 89.1 on DAVIS 2017): distilled the image encoder into MobileNetV4 (3.6× fewer parameters) and, after profiling showed memory attention dominated each frame, distilled it to attend to 3 memory frames instead of 7. MLflow, ONNX Runtime, CI with an eval gate, FastAPI + Docker.
- **arXiv RAG: question answering with citations.** Hybrid retrieval (dense + BM25 fused with RRF, then a cross-encoder reranker) and a local LLM that answers only from cited sources or abstains. A LangGraph corrective-RAG agent fetches and indexes new arXiv papers when the index can't answer. On 200 questions with 95% bootstrap confidence intervals, raised MRR from 0.45 to 0.71 and Hit@5 from 0.61 to 0.82 vs. dense-only retrieval. LangChain, FastAPI, Docker.
- **Real-time face and emotion detection.** Main contributor in a team of 3. Webcam app chaining two fine-tuned YOLOv8 models: a face detector (mAP50 0.89) feeding an 8-class emotion model trained on AffectNet (mAP50 0.80).
- 2025 **Night on Fire: game jam.** Programmer in a team of 5 at SummerJam 2025 (Granada); game built in Godot during the jam, playable on itch.io.
- **Battle Royale Optimization (BRO).** Implemented the BRO metaheuristic and a memetic variant that applies local search to the population to balance exploration and exploitation; benchmarked on CEC2017.

Skills

- Programming: Python, C++, R, SQL
- Backend: FastAPI, Docker, Gradio
- Databases: MySQL, SQLite, ChromaDB
- Tooling: Git, GitHub Actions (CI), pytest, uv, Ruff
- ML / DL: PyTorch, scikit-learn, fastai, Ollama, LangChain, LangGraph, pandas, NumPy, Matplotlib
- MLOps: MLflow, ONNX
- Languages: Spanish (native), English (C1, advanced)

Soft skills

- Analytical thinking, Problem solving, Teamwork, Adaptability, Technical communication, Creativity, Time management, Initiative, Continuous learning.